Indian Institute of Technology Kanpur

Department of Space, Planetary & Astronomical Sciences & Engineering (SPASE)

SPA617: Radio Astronomy Laboratory

RF Components: Characterization and Measurement part-2

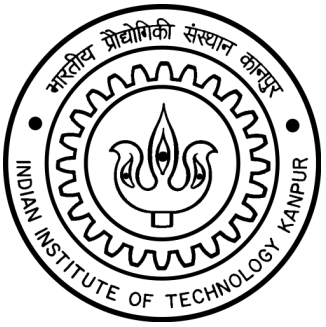
Laboratory Report

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Aim and the theory already have given in the part 1, here we just provide the rest of part of ourexperiment.

A2. Filter Input Reflection Coefficient and Return Loss

The input reflection coefficient of the filter was measured using the maximum and minimum voltage amplitudes at the Tee junction.

The measured values were

$$f = 68 \text{ MHz}, \tag{1}$$

$$V_{\max} = 1.75 \text{ V}, \qquad V_{\min} = 987 \text{ mV} = 0.987 \text{ V}. \tag{2}$$

According to the measurement procedure, the incident and reflected voltage amplitudes are

$$V_{\text{inc}} = \frac{V_{\max} + V_{\min}}{2}, \tag{3}$$

and

$$V_{\text{ref}} = \frac{V_{\max} - V_{\min}}{2}. \tag{4}$$

Therefore,

$$V_{\text{inc}} = \frac{1.75 + 0.987}{2} \tag{5}$$

$$= 1.3685 \text{ V}, \tag{6}$$

and

$$V_{\text{ref}} = \frac{1.75 - 0.987}{2} \tag{7}$$

$$= 0.3815 \text{ V}. \tag{8}$$

The magnitude of the reflection coefficient is

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} \tag{9}$$

$$= \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{10}$$

$$= \frac{1.75 - 0.987}{1.75 + 0.987} \tag{11}$$

$$\approx \boxed{0.279}. \tag{12}$$

The corresponding return loss is

$$RL = -20 \log_{10} |\Gamma| \tag{13}$$

$$= -20 \log_{10}(0.279) \tag{14}$$

$$\approx \boxed{11.1 \text{ dB}}. \tag{15}$$

A2 Result

At 68 MHz, the measured magnitude of the input reflection coefficient was approximately

$$\boxed{|\Gamma| \approx 0.279}, \tag{16}$$

with a corresponding return loss of

$$\boxed{RL \approx 11.1 \text{ dB}}. \tag{17}$$

C2. Power Combiner

The following measurements were taken over the frequency range from 50 to 200 MHz.

The measured output powers are listed in Table 1.

Table 1: Measured output power of the RF power combiner at different frequencies.

Frequency (MHz)	Output power (dBm)
50	-2.89
100	-3.20
150	-3.12
200	-3.68

Observation

For equal-frequency inputs, the combined output contains a signal at the common input frequency. and power behaves as per the table. For different-frequency inputs, the output contains the two input frequency components. Since the device is a passive power combiner, sum and difference frequency components are not expected as they would be in a nonlinear mixer.

PART D: AMPLIFIER CHARACTERIZATION

Part D of the RF Components experiment involved the characterization of an RF power amplifier. The amplifier was supplied using a DC power supply, while an RF signal generator was used to provide the input signal. The amplifier output was connected to a spectrum analyser for measuring the output power. The amplifier was characterized in three stages. First, the gain was measured over a range of frequencies while keeping the input power approximately constant. Second, the input power was increased at a fixed frequency to investigate gain compression and determine the 1-dB compression point. Finally, a two-tone input signal was applied to investigate third-order intermodulation products and determine the third-order intercept point (TOI).

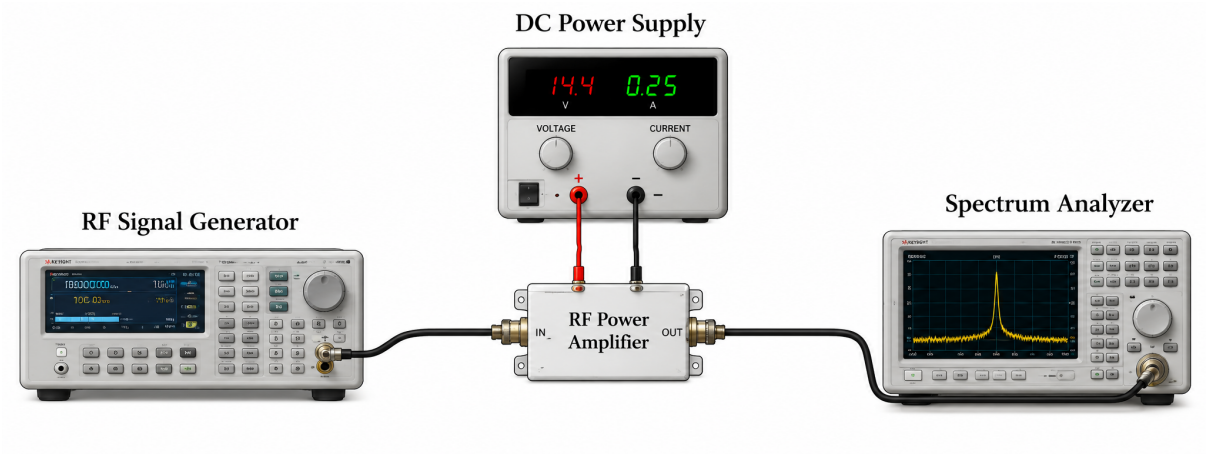


Figure 1: Experimental setup used for RF power amplifier characterization.

D1. Gain versus Frequency

For this measurement, the amplifier input power was kept constant at

$$P_{\text{in}} = -20 \text{ dBm}. \tag{18}$$

The output power was measured over the operating frequency range. The amplifier gain was calculated using

$$G = P_{\text{out}} - P_{\text{in}}. \tag{19}$$

The measured output power and calculated gain are given in Table 2. For example, at 100 MHz,

$$\begin{aligned} G &= P_{\text{out}} - P_{\text{in}} && (20)\\ &= 2.40 - (-20) && (21)\\ &= \boxed{22.40 \text{ dB}}. && (22) \end{aligned}$$

Similarly, at 600 MHz,

$$\begin{aligned} G &= 0.46 - (-20) && (23)\\ &= \boxed{20.46 \text{ dB}}. && (24) \end{aligned}$$

Table 2: Measured amplifier output power and gain as a function of frequency.

Frequency (MHz)	P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
100	-20	2.40	22.40
150	-20	2.07	22.07
200	-20	1.98	21.98
250	-20	1.68	21.68
300	-20	1.70	21.70
350	-20	1.40	21.40
400	-20	1.27	21.27
450	-20	1.21	21.21
500	-20	0.62	20.62
550	-20	0.77	20.77
600	-20	0.46	20.46

The measured gain decreases gradually with increasing frequency. The gain-versus-frequency response is shown in Fig. 2.

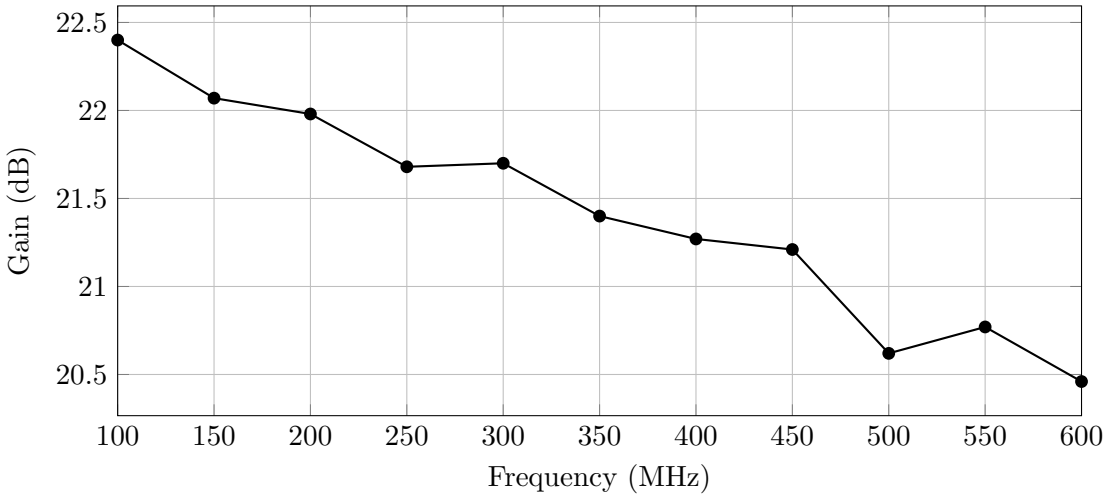


Figure 2: Measured amplifier gain versus frequency.

D1 Result

The measured amplifier gain was 22.40 dB at 100 MHz and decreased to 20.46 dB at 600 MHz. Thus, the amplifier showed a gradual decrease in gain with frequency over the measured range.

D2. Harmonics and 1-dB Compression Point

The compression measurement was performed at

$$f = 100 \text{ MHz}. \tag{25}$$

The input power was increased gradually and the corresponding output power was measured. The amplifier gain was calculated using

$$G = P_{\text{out}} - P_{\text{in}}. \tag{26}$$

The measured results are given in Table 3.

P_{in} (dBm)	P_{out} (dBm)	Gain (dB)
-20	2.40	22.40
-19	3.30	22.30
-18	4.37	22.37
-17	5.34	22.34
-16	6.25	22.25
-15	7.34	22.34
-14	8.30	22.30
-13	9.20	22.20
-12	10.20	22.20
-11	11.29	22.29
-10	12.20	22.20
-9	13.20	22.20
-8	14.06	22.06
-7	14.85	21.85
-6	15.50	21.50
-5	16.02	21.02
-4	16.40	20.40

Table 3: Amplifier output power and gain as a function of input power at 100 MHz.

The reference gain was taken from the maximum amplifier gain measured at 100 MHz in D1:

$$G_{\text{max}} = 22.40 \text{ dB}. \tag{27}$$

The gain corresponding to a 1-dB reduction from this reference value is therefore

$$\begin{aligned} G_{\text{1dB}} &= G_{\text{ref}} - 1 && (28)\\ &= 22.40 - 1 && (29)\\ &= \boxed{21.40 \text{ dB}}. && (30) \end{aligned}$$

From the measured D2 data, the gain is 21.50 dB at $P_{\text{in}} = -6$ dBm and 21.02 dB at $P_{\text{in}} = -5$ dBm. Hence, the 21.40-dB level lies between these two measured points.

Linear interpolation gives

$$\begin{aligned} P_{\text{in,1dB}} &= -6 + \frac{21.50 - 21.40}{21.50 - 21.02}(1) && (31)\\ &= -5.79 \text{ dBm}. && (32) \end{aligned}$$

The corresponding output power is obtained by interpolation between 15.50 and 16.02 dBm:

$$\begin{aligned} P_{\text{out,1dB}} &= 15.50 + \frac{21.50 - 21.40}{21.50 - 21.02}(16.02 - 15.50) && (33)\\ &= 15.61 \text{ dBm}. && (34) \end{aligned}$$

Therefore, the measured and interpolated 1-dB compression point is

$$\boxed{P_{\text{in,1dB}} \approx -5.79 \text{ dBm}} \tag{35}$$

with corresponding output power

$$\boxed{P_{\text{out,1dB}} \approx 15.61 \text{ dBm}.} \tag{36}$$

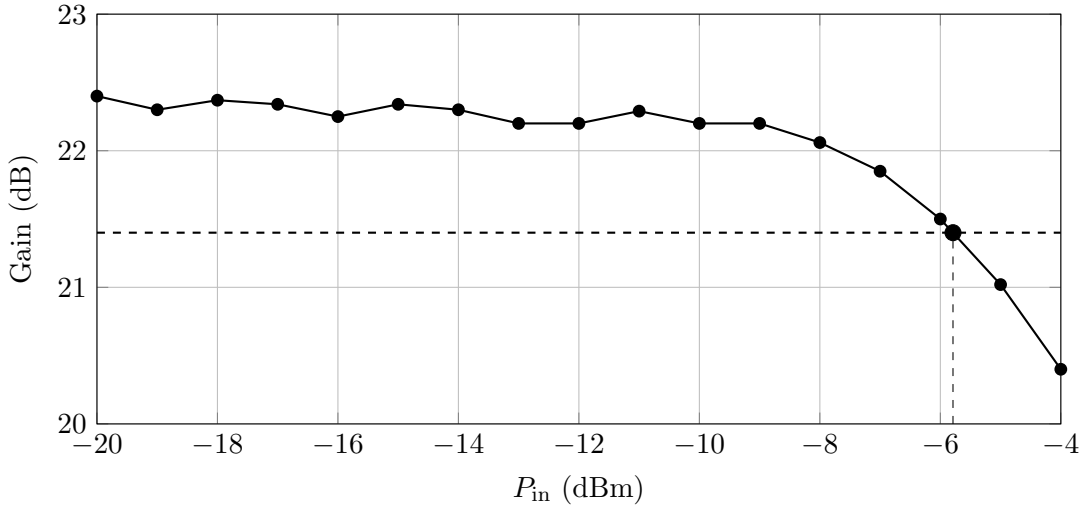


Figure 3: Measured amplifier gain versus input power at 100 MHz. The horizontal dashed line represents the 1-dB-reduced gain level, and the highlighted point indicates the interpolated 1-dB compression point.

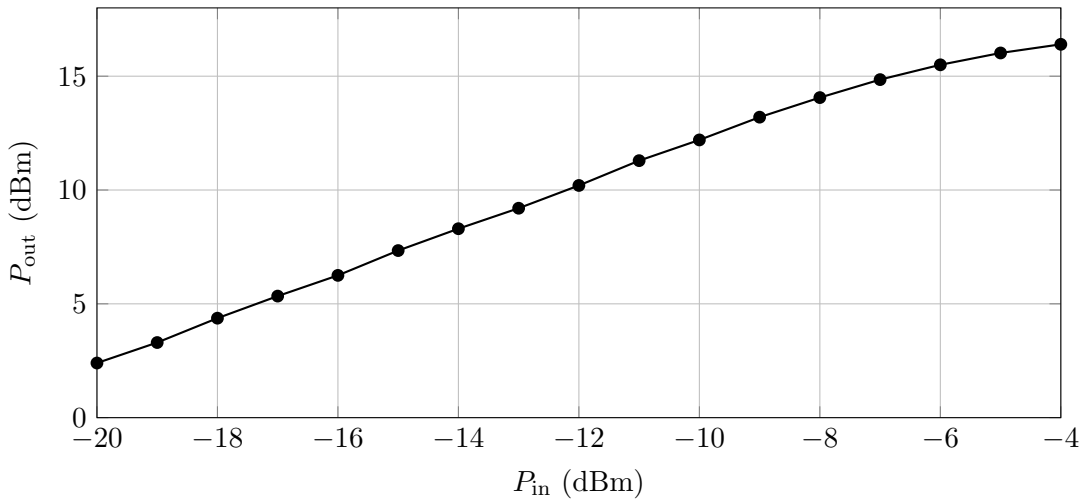


Figure 4: Measured amplifier output power as a function of input power at 100 MHz.

The variation of amplifier output power with input power was measured at 100 MHz. The measured P_{out}–P_{in} characteristic is shown in Fig. 4. At low input power, the output increases approximately linearly with input power. At higher input power, the curve begins to deviate from the linear behaviour, indicating the onset of gain compression.

D2 Result

The amplifier showed nearly constant gain at low input power, indicating approximately linear small-signal operation. The maximum measured gain at 100 MHz was 22.40 dB. A 1-dB reduction from this value corresponds to a gain of

$$G_{1\text{dB}} = 22.40 - 1.00 = 21.40 \text{ dB}. \tag{37}$$

As the input power was increased, the amplifier gain gradually decreased due to compression. The gain of 21.40 dB lies between the measured values of 21.50 dB at $P_{\text{in}} = -6$ dBm and 21.02 dB at $P_{\text{in}} = -5$ dBm. Linear interpolation between these two measurements gives

$$\boxed{P_{\text{in,1dB}} \approx -5.79 \text{ dBm}} \tag{38}$$

with a corresponding output power of

$$\boxed{P_{\text{out,1dB}} \approx 15.61 \text{ dBm}}. \tag{39}$$

The 1-dB compression point therefore provides an estimate of the power level at which noticeable gain compression begins.

D3. Third-Order Intercept (TOI)

The third-order intercept point was investigated using a two-tone signal. The two input frequencies were

$$f_1 = 10 \text{ MHz}, \qquad f_2 = 13 \text{ MHz}. \tag{40}$$

For a nonlinear amplifier, third-order intermodulation products occur at

$$2f_1 - f_2 \tag{41}$$

and

$$2f_2 - f_1. \tag{42}$$

For the selected frequencies,

$$\begin{aligned} 2f_1 - f_2 &= 2(10) - 13 && (43)\\ &= 7 \text{ MHz}, && (44) \end{aligned}$$

and

$$\begin{aligned} 2f_2 - f_1 &= 2(13) - 10 && (45)\\ &= 16 \text{ MHz}. && (46) \end{aligned}$$

The measured fundamental output power and third-order product power for different input powers per tone are given in Table 4.

Table 4: Two-tone amplifier measurement for TOI determination.

P_{in} per tone (dBm)	P_{out} fundamental (dBm)	P_{3rd} (dBm)
-10	8.75	-27.62
-9	9.66	-24.02
-8	10.78	-20.14
-7	11.44	-17.34
-6	12.25	-14.46
-5	13.08	-11.80
-4	13.83	-9.43
-3	14.54	-7.32
-2	15.10	-5.60
-1	15.70	-4.20
0	16.13	-3.39
+1	16.54	-3.01

At low input powers, the fundamental output increases approximately linearly with input power, whereas the third-order product increases much more rapidly. Ideally, the fundamental component has a slope of approximately 1 dB/dB, while a third-order product has a slope of approximately 3 dB/dB.

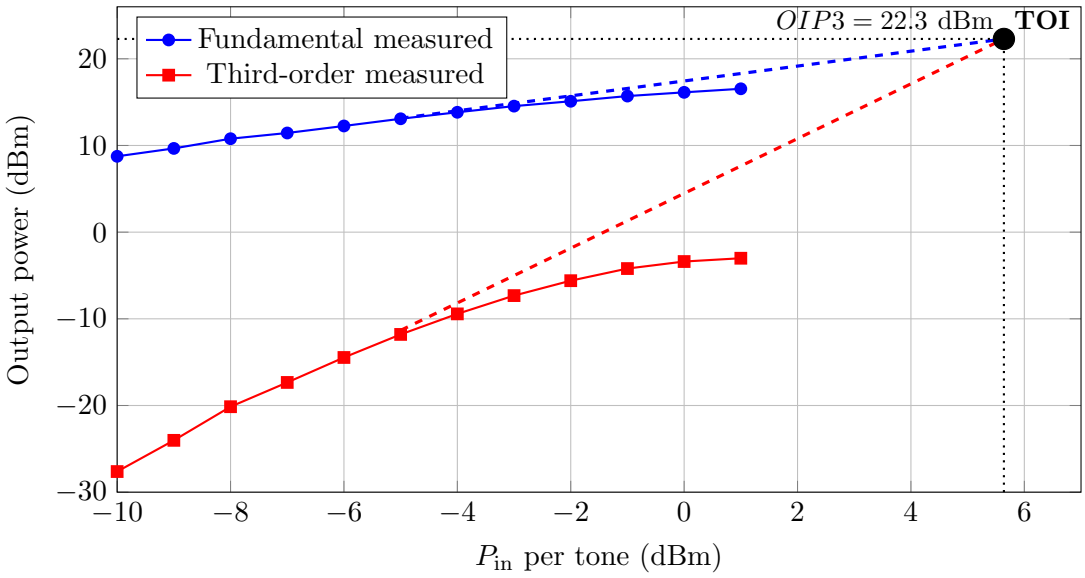


Figure 5: Measured fundamental and third-order intermodulation output powers as a function of input power per tone. The dashed lines represent the extrapolated fundamental and third-order trends. Their intersection gives the third-order intercept point (TOI), from which $IIP3$ and $OIP3$ are obtained.

For the TOI determination, the low-input-power region was used to obtain the approximately linear trends of the fundamental and third-order responses. The fundamental response was fitted as

$$P_{\text{out}} \approx 0.86P_{\text{in}} + 17.44, \tag{47}$$

while the third-order response was fitted as

$$P_{\text{3rd}} \approx 3.16P_{\text{in}} + 4.47. \tag{48}$$

The TOI occurs where the extrapolated fundamental and third-order responses become equal. Therefore,

$$0.86P_{\text{in}} + 17.44 = 3.16P_{\text{in}} + 4.47. \tag{49}$$

Hence,

$$P_{\text{in,TOI}} = \frac{17.44 - 4.47}{3.16 - 0.86} \tag{50}$$

$$\approx 5.64 \text{ dBm}. \tag{51}$$

The corresponding output power is

$$P_{\text{out,TOI}} = 0.86(5.64) + 17.44 \tag{52}$$

$$\approx 22.3 \text{ dBm}. \tag{53}$$

Thus,

$$\boxed{IIP3 \approx 5.64 \text{ dBm}} \tag{54}$$

and

$$\boxed{OIP3 \approx 22.3 \text{ dBm}}. \tag{55}$$

D3 Result

The measured fundamental and third-order intermodulation powers are shown in Fig. 5. The extrapolated fundamental and third-order trends intersect at an input power of approximately

$$\boxed{IIP3 \approx 5.64 \text{ dBm}}. \tag{56}$$

The corresponding output power at the intersection is

$$\boxed{OIP3 \approx 20.43 \text{ dBm}}. \tag{57}$$

So, the two-tone measurement indicates measurable third-order intermodulation distortion, with the TOI obtained by extrapolating the measured fundamental and third-order responses beyond the experimental input-power range.

Part 2 Conclusion

The RF power amplifier was characterized in terms of its frequency response, gain compression, and third-order intermodulation distortion. The amplifier showed a maximum measured gain of 22.40 dB at 100 MHz, which gradually decreased with frequency to 20.46 dB at 600 MHz.

At 100 MHz, the amplifier exhibited nearly constant gain at low input powers. With increasing input power, the gain decreased, indicating the onset of gain compression. The 1-dB compression point was determined using the maximum measured gain, giving

$$P_{\text{in,1dB}} \approx -5.79 \text{ dBm}, \qquad P_{\text{out,1dB}} \approx 15.61 \text{ dBm}. \tag{58}$$

The two-tone measurement demonstrated third-order intermodulation distortion. For input frequencies of 10 MHz and 13 MHz, the third-order products were observed at 7 MHz and 16 MHz. Extrapolation of the measured fundamental and third-order responses gave

$$IIP3 \approx 5.64 \text{ dBm}, \qquad OIP3 \approx 20.43 \text{ dBm}. \tag{59}$$

Main error due to intrument handling and proper working.

Use of AI Tools

Artificial intelligence tools were used as an assistance tool during the preparation of this laboratory report. The use of AI to support the preparation, checking, and presentation of the work. The main uses were:

- **Data organization:** To organize the experimentally recorded data into clear tables and suitable formats for analysis.
- **Numerical calculations:** To perform and verify calculations such as amplifier gain, the 1-dB compression point, third-order intercept point (TOI), and other derived quantities.

- **LaTeX writing and formatting:** To convert equations, tables, figures, captions, and other report content into properly formatted LaTeX code suitable for Overleaf.
- **Language improvement:** To improve the grammar, sentence structure, clarity, and technical wording of the report.
- **Report structure:** To organize the report into a logical structure, including experimental measurements, calculations, tables, plots, results, and conclusions.
- **Plots and figures:** To prepare and format plots from the experimentally recorded data and to assist in presenting the measured results clearly.
- **Checking and verification:** To check numerical calculations, equations, tables, and consistency between the measured data and the reported results.
- **Presentation of results:** To improve the overall readability and presentation of the experimental results without changing the experimentally obtained measurements.